Chi-Sheng (Michael) Chen (陳麒升)

Machine Learning for Physiological Time Series | Clinical Biosignals & Foundation Models | EEG/BCI | Quantum ML

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Nationality: Taiwan | Based in Cambridge, MA, USA

Chi-Sheng (Michael) Chen builds machine learning for physiological time series along one arc — from representation, to foundation models, to clinical deployment. He develops interpretable, clinician-interrogable frequency-domain and geometric representations of clinical biosignals (FreqLens, SPD Token Transformer) and scales them into biosignal foundation models (Large Cognition Model). His EEG-based depression-treatment model is in routine outpatient use at Taipei Veterans General Hospital's Precision Depression Intervention Center (PreDIC), and at Harvard/BIDMC he builds multimodal EMS trauma-triage pipelines. He has 49 publications, 515 citations (h-index 13, i10-index 16), ICML 2026 Gold Reviewer recognition, and reviewer service across 46 journals (incl. TPAMI, NSR, npj Quantum Information). As a secondary methodological line, he explores hybrid quantum-classical architectures for sequence modeling (QASA, QEEGNet).

Research & Professional Experience

Harvard Medical School & Beth Israel Deaconess Medical Center — MA, USA

Research Affiliate (primary appointment), Advisor: Prof. Dr. Gabriel Brat — *Nov 2024 – Present*

- Building a real-time EMS trauma-triage pipeline (ASR → LLM-agent event extraction → structured prediction) over **4M+ patient records** using multimodal AI.
- Co-developed webcam-based open-surgical skill assessment (hand keypoint detection; CLINICCAI 2026 poster) and a capture-clock methodology for browser webcam gaze (DEMI 2026 spotlight talk).

WhaleForce (subsidiary of VICI Holdings) — Taipei, Taiwan

Fractional Token Efficiency Research Lead (Advisor) — *Aug 2026 – Present*

- Advising on LLM token-efficiency research and multi-agent architecture for AI-driven quantitative trading.

Omnis Labs — Remote

Co-Founder — *Sep 2024 – May 2026*

- AI-driven DeFi liquidity positioning protocol with automated market maker (AMM) strategies and backtesting systems.
- Achieved 50%–120% base fee APR on WBTC/USDC and ETH/USDC pairs (Katana on SushiSwap) using time-series model-driven AMM strategies.

Neuro Industry, Inc. — CA, USA

Researcher, Co-Founder & CTO — *Mar 2024 – Jan 2025*

- Built GCP-based MLOps pipeline, enabling training on 10K+ EEG samples.
- 3 research papers: quantum machine learning for EEG signal processing (2) and EEG-to-image diffusion (1).

Department of Computer Science, National Yang Ming Chiao Tung University — Hsinchu, Taiwan

Research Assistant, Advisor: Prof. Chun-Shu Wei — *Dec 2023 – Aug 2024*

- Developed MUSE, a multimodal contrastive learning framework for EEG-based image recognition (34 citations).
- Research on nonstationary EEG dynamics and foundation models for brain-computer interfaces.

MediaTek Inc. — Hsinchu, Taiwan

Digital IC R&D Engineer — *Dec 2021 – Oct 2023*

- Responsible for microprocessor IP development for flagship 5G smartphones' display and AMBA SoC implementation.
- Hardware virtualization architecture RTL design and IP verification with UVM and SystemVerilog.

Department of Surgery, National Taiwan University Hospital — Taipei, Taiwan

Research Assistant, Advisor: Dr. Shuo-Lun Lai — *Jul 2021 – Sep 2021*

- Deep learning applications on surgery automation with YOLO-based model.
- Developed a prototype for a surgical smoke evacuation system.

Department of Psychiatry, Taipei Veterans General Hospital — Taipei, Taiwan

Research Assistant, Advisor: Prof. Dr. Cheng-Ta Li — *Sep 2019 – Jun 2021*

- Collaborated with researchers and clinicians to identify patterns and trends in brainwave activity related to major depressive disorder.
- Utilized machine learning algorithms to develop predictive models based on real-world brainwave data.

Max Planck Institute for Chemical Physics of Solids (MPI CPfS) — Dresden, Germany

Research Internship, Advisor: Dr. Alexander Komarek & Dr. Li Zhao — *Jul 2018 – Sep 2018*

- Searching new possible unconventional superconductors among Co-based quaternary chalcogenides with diamond like structure $CuInCo_2A_4$ / $AgInCo_2A_4$ (A = Te, Se, S).

Other: OpenAI (contractor via Mercor) — AI Trainer, RLHF preference data & model-behavior evaluation *Mar 2025 – Oct 2025*

Education

National Taiwan University (NTU) Taipei, Taiwan
Master of Science in Biomedical Electronics and Bioinformatics *Sep 2019 – Jun 2021*
Advisor: Prof. Dr. Cheng-Ta Li, MD & Prof. Chung-Ping Chen

- Machine learning application on non-stationary time series data.
- The research EEG model for TMS pattern personalized prediction is currently being applied to real outpatient patients in the Precision Depression Intervention Center (PreDIC), Taipei Veterans General Hospital.
- Master's thesis: "EEG Analysis for Prediction of Antidepressant Responses of Transcranial Magnetic Stimulation in Major Depressive Disorder Based on Attentional Convolution Time Series Prototypical Neural Network Model and Classical/Quantum Machine Learning Approaches."

National Chiao Tung University (NCTU) Hsinchu, Taiwan
Bachelor of Engineering in Electrophysics *Sep 2015 – Jun 2019*
Bachelor of Science in Interdisciplinary Science Degree Program

- Researching the oxide processing techniques.
- Researching the biomedical signal processing on traditional Chinese medicine data.

Skills

AI/ML: Python, PyTorch, HuggingFace, Scikit-learn, MNE, MLflow, R, MATLAB
Systems: C/C++, JavaScript, Rust, MySQL, Solidity, Git, Docker, GCP, AWS
Hardware: Verilog, SystemVerilog, UVM, HSPICE, Genus, Verdi, SG-Lint, Perl
Languages: Chinese (Native), English (Fluent)

Selected Publications (515 cites, h-index 13, i10-index 16)

First author unless noted. Peer-reviewed unless marked *Under review* or arXiv preprint.
[Physiological Time Series & Clinical Biosignals]

Chen, C.S., et al. "FreqLens: Interpretable Frequency Attribution for Time Series Forecasting" *Under review*, ICLR 2027; arXiv:2602.08768.

Chen, C.S., et al. "A Unified SPD Token Transformer Framework for EEG Classification: Systematic Comparison of Geometric Embeddings" *Under review*, Neurocomputing (Elsevier), 2026; arXiv:2601.21521.

Chen, C.S., et al. "Large Cognition Model: Towards Pretrained EEG Foundation Model" arXiv preprint (2025), arXiv:2502.17464 — **30 citations**.

Chen, C.S., Wei, C.S. "Mind's Eye: Image Recognition by EEG via Multimodal Similarity-Keeping Contrastive Learning" arXiv preprint (2024), arXiv:2406.16910 — **34 citations**.

Chen, C.S., et al. "Exploring the Potential of Electroencephalography Signal–Based Image Generation Using Diffusion Models: Integrative Framework Combining Mixed Methods and Multimodal Analysis" JMIR Medical Informatics (2025), https://doi.org/10.2196/72027 .

Chen, C.S., et al. "Exploring the potential of QEEGNet for cross-task and cross-dataset electroencephalography encoding with quantum machine learning" Journal of Signal Processing Systems 98, 5 (2026), https://doi.org/10.1007/s11265-025-01979-2 .

Chen, C.S., et al. "QEEGNet: Quantum Machine Learning for Enhanced Electroencephalography Encoding" IEEE International Workshop on Signal Processing Systems (SiPS) 2024, doi: 10.1109/SiPS62058.2024.00035, arXiv:2407.19214; poster at QTML 2024, Melbourne (non-archival).

Chen, C.S., Brat, G.A. "Measuring Browser Webcam Gaze Honestly: A Capture-Clock Methodology and Open Reference Implementation" 4th MICCAI Workshop on Data Engineering in Medical Imaging (DEMI 2026), Spotlight talk, arXiv:2608.11566.

Li, C.T., **Chen, C.S., et al. "Prediction of antidepressant responses to non-invasive brain stimulation using frontal electroencephalogram signals: Cross-dataset comparisons and validation"** Journal of Affective Disorders (2023), https://doi.org/10.1016/j.jad.2023.08.059 . [2nd author; model deployed at Taipei VGH PreDIC]

[Quantum Machine Learning]

Chen, C.S., Kuo, E.J. "Quantum Adaptive Self-Attention for Quantum Transformer Models" Quantum Science and Technology (2026), https://doi.org/10.1088/2058-9565/ae8b21 .

Chen, C.S., et al. "Unraveling quantum environments: Transformer-assisted learning in Lindblad dynamics" Physical Review A (2025), https://doi.org/10.1103/gsxk-45mk .

Chen, C.S., et al. "Quantum Reinforcement Learning-Guided Diffusion Model for Image Synthesis via Hybrid Quantum-Classical Generative Model Architectures" 2026 IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP), doi: 10.1109/ICASSP55912.2026.11461991 .

Chen, C.S., et al. "Quantum Adaptive Self-Attention for Financial Rebalancing: An Empirical Study on Automated Market Makers in Decentralized Finance" 2026 IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP), doi: 10.1109/ICASSP55912.2026.11460972 .

Chen, C.S., et al. "Quantum Multimodal Contrastive Learning Framework" 2025 IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP), doi: 10.1109/ICASSP49660.2025.10889504, arXiv:2408.13919 .

Chen, C.S., et al. "Q-DPTS: Quantum Differentially Private Time Series Forecasting via Variational Quantum Circuits" IEEE GLOBECOM Workshop 2025, arXiv:2508.05036.

Chen, C.S., et al. "Quantum Reinforcement Learning Trading Agent for Sector Rotation in the Taiwan Stock Market" IEEE QCE QCRL Workshop 2025, arXiv:2506.20930.

[Computer Vision, Digital Humanities & Physics]

Chen, C.S., et al. "Improving fine-grained food classification using deep residual learning and selective state space models" PLoS ONE (2025), https://doi.org/10.1371/journal.pone.0322695 .

Chen, C.S., Liu, H.Y. "Predicting Poets' Origins from Verse: A Computational Analysis of Regional Linguistic Fingerprints in the Complete Tang Poems" Digital Humanities 2026 (DH2026), Alliance of Digital Humanities Organizations (ADHO), Short Paper.

Lai, S.L., **Chen, C.S., et al. "Intraoperative Detection of Surgical Gauze Using Deep Convolutional Neural Network"** Ann Biomed Eng 51, 352–362 (2023), https://doi.org/10.1007/s10439-022-03033-9 . [2nd author]

Lai, S.L., Chou, Y.C., **Chen, C.S., et al. "Detection and tracking of a gauze sponge in minimally invasive surgery using a YOLO and R-CNN based model"** Medical & Biological Engineering & Computing (2025), https://doi.org/10.1007/s11517-025-03471-2 . [3rd author]

Liu, H.Y., **Chen, C.S.** "Gendered Language in Tang Poetry: A Sociolinguistic and Digital Humanities Analysis Based on the Quan Tang Shi." 16th International Conference of Digital Archives and Digital Humanities (DADH 2025).

Takegami D., et al. **"Direct imaging of valence orbitals using hard x-ray photoelectron spectroscopy"** Phys. Rev. Res. 4, 033108 (2022), https://doi.org/10.1103/PhysRevResearch.4.033108 .

Falke J., et al. **"Electronic structure of the metallic oxide ReO_3"** Phys. Rev. B 103, 115125 (2021), https://doi.org/10.1103/PhysRevB.103.115125 .

Plus 18 further preprints across quantum machine learning, time-series forecasting and medical imaging (full list: Google Scholar).

Invited Talks & Presentations

Guest Lecture Series, Introduction to Biomedical Engineering, Dept. of Biomedical Engineering, Chang Gung University (host: Prof. Shao-Hsuan Chang) — Biosignal Processing, three sessions, Oct–Nov 2026.

Guest Lecture, Wearable Chip Design, NYCU ECE (host: Prof. Sheng-Chieh Huang) — "Quantum Machine Learning for Wearable Biosignals," Oct 2026.

DEMI 2026 (4th MICCAI Workshop on Data Engineering in Medical Imaging), Strasbourg, France — spotlight talk, Oct 2026.

CLINICCAI 2026 (Clinical Day at MICCAI 2026), Strasbourg, France — poster, presenting author, Sep 2026.

Guest Lecture, Maker Club, NYCU, Hsinchu — "AI, Robotics and Makers in Agent Era," Sep 2026.

Guest Lecture, Advanced Compilers, NTU CSIE (host: Prof. Shih-Wei Liao) — "Introduction to Classical/Quantum Biosignal Processing," Sep 2026.

Digital Humanities 2026 (DH2026), Daejeon, South Korea — two short-paper presentations, Jul 2026.

IEEE ICASSP 2026, Barcelona, Spain — two oral presentations, May 2026.

NVIDIA GTC 2026, San Jose, CA — two poster presentations, Mar 2026 (**Top 8 Finalist**).

Invited Seminar, Dept. of Applied Physics, Tunghai University, Taichung — Dec 2025.

IEEE Quantum Week (QCE 2025), Albuquerque, NM — paper presentation, Aug 2025.

IEEE ICASSP 2025, Hyderabad, India — oral presentation, Apr 2025.

QTML 2024 (8th Int. Conf. on Quantum Techniques in Machine Learning), Melbourne, Australia — poster, QEEGNet, Nov 2024.

Professional Service

Program Committee: QNRL Workshop @ IEEE WCCI 2026

Conference Reviewer (2026–2027): **NeurIPS**, **ICML** (Gold Reviewer — top tier), **AAAI**, **KDD**, **MICCAI**, **ICASSP**; NLDL 2027; ACM ICAIF 2026 (7th Int. Conf. on AI in Finance); IEEE QAI 2026

IOP Trusted Reviewer (2026), recognized by the Institute of Physics Publishing for quality and commitment in peer review.

Journal Reviewer (2025–present) — **46 journals**, incl. IEEE TPAMI, Oxford National Science Review, IEEE TNNLS, Elsevier Neural Networks, IEEE/ACM TASLP, IEEE JBHI, IEEE TNSRE, IEEE TQE, npj Quantum Information, Scientific Reports, ACM TORS (full list: chishengchen.github.io).

Awards & Achievements

2026 NVIDIA GTC Top 8 Finalist: "GPU-Accelerated Chinese Brain-to-Text Interface Powered by NVIDIA H100 80GB."

2026 Berkeley RDI AgentX-AgentBeats: Phase 1 Healthcare Agent 3rd Place (Team MadGAA, 1,300+ teams worldwide).

2023 Certificate of the 20th National Innovation Award (Taiwan), clinical research category — for the EEG treatment-response model now deployed at Taipei VGH PreDIC.

2019 International Blockchain Olympiad (IBCOL): World Finalist competition with 100+ teams.

2017 Microsoft International Imagine Cup: 2nd place in Taiwan.

2016 International Genetically Engineered Machine Competition (iGEM): World Gold medal, Best Applied Design, Best Part Collection, Best Presentation competition with 300+ university teams.